

Do Tran Hieu

DATA ENGINEER INTERN

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Summary

Data Engineer Intern with strong interest in building scalable data pipelines, ETL workflows, and analytical data systems. Experienced in designing data warehouses, orchestrating workflows with Apache Airflow, and working with columnar databases such as ClickHouse. Familiar with ML pipelines and data visualization using Power BI.

Education

Posts and Telecommunications Institute of Technology (PTIT)

2023 – Present

Major: Computer Science

GPA: 3.44 / 4.0 CPA: 3.47 / 4.0

Honors & Awards

Academic Excellence Scholarship (Excellent Category)

Semester 4, 2025 – 2026

Awarded by PTIT for outstanding academic performance (GPA-based merit scholarship).

Entrance Scholarship

The first year 2023 - 2024

Awarded Entrance Scholarship based on academic excellence.

English Certification

Toeic 745 (2025 - 2027)

Skills

Data Engineering: ETL Pipeline Design, Data Warehouse (Star Schema), Data Modeling, Apache Airflow

Databases & Object Storage: ClickHouse (OLAP), MySQL, PostgreSQL, MinIO, Google BigQuery

Programming: Python, SQL, Java

Big Data & Tools: Pandas, Jupyter, Docker, Kafka (basic)

Machine Learning: Feature Engineering, XGBoost

Tools: Git, GitLab, Power BI

Projects

Football Data Warehouse: ETL, ML & Analytics System

May 2026 - June 2026

Github: <https://github.com/tranhieu2005do/HQTCSDL>

- Built an end-to-end data engineering pipeline: Extract → Transform → Load (ETL) from API-Football into ClickHouse.
- Designed a scalable **Star Schema Data Warehouse** with dimension and fact tables optimized for OLAP queries.
- Implemented automated data orchestration using **Apache Airflow (Dockerized)** with DAG scheduling, validation layers, and short-circuit failure handling.
- Developed data transformation pipelines using **Pandas + Jupyter/Papermill**, including data cleaning, normalization, and feature generation.
- Engineered advanced football features such as: recent form, head-to-head stats, rest days, home/away performance metrics.
- Built **ML pipeline using XGBoost + Optuna** for match score prediction with time-series split validation.
- Designed Power BI dashboards for performance analytics: league ranking, xG analysis, player statistics.

Core Strengths

- End-to-end ETL pipeline design and implementation
- Data warehouse modeling (Star Schema, OLAP optimization)
- Workflow orchestration with Apache Airflow
- ML feature engineering for structured data
- Distributed data processing mindset (Dockerized pipelines)

Interests

Data Engineering, Data Warehousing, Distributed Systems, Machine Learning